Check for updates

AGU ADVANCING EARTH AND SPACE SCIENCE

Geophysical Research Letters®

RESEARCH LETTER
10.1029/2021GL096902

**Key Points:**

- Volcanic rocks of the Late Triassic Batang Group are dated by U-Pb on zircon to ~227–222 Ma
- Paleomagnetic data from these rocks yield a robust Late Triassic paleopole of 57.6°N, 176.4°E($A_{95}$ = 7.8°)
- Closure of the Paleo-Jinshajiang Ocean occurred before 227–222 Ma, and likely around ~230 Ma

**Supporting Information:**

Supporting Information may be found in the online version of this article.

**Correspondence to:**

M. Yan,
maoduyan@itpcas.ac.cn

**Citation:**

Yu, L., Yan, M., Domeier, M., Guan, C., Shen, M., Fu, Q., et al. (2022). New paleomagnetic and chronological constraints on the Late Triassic position of the Eastern Qiangtang Terrane: Implications for the closure of the Paleo-Jinshajiang Ocean. *Geophysical Research Letters*, *49*, e2021GL096902. https://doi.org/10.1029/2021GL096902

Received 4 NOV 2021
Accepted 4 JAN 2022

**Author Contributions:**

**Conceptualization:** Maodu Yan, Mathew Domeier, Chong Guan, Miaomiao Shen, Qiang Fu, Rendeng Shi, Weilin Zhang
**Data curation:** Liang Yu, Mathew Domeier, Chong Guan, Zunbo Xu, Zhichao Niu, Liye Yang, Jinbo Zan, Dawen Zhang, Bingshuai Li
**Formal analysis:** Liang Yu, Maodu Yan, Mathew Domeier, Qiang Fu, Zunbo Xu, Weilin Zhang, Jinbo Zan
**Funding acquisition:** Maodu Yan
**Investigation:** Liang Yu, Maodu Yan, Chong Guan, Qiang Fu, Wanlong Xu, Liye Yang, Weilin Zhang, Bingshuai Li
**Methodology:** Liang Yu, Maodu Yan, Mathew Domeier, Chong Guan, Miaomiao Shen, Qiang Fu, Wanlong Xu, Zhichao Niu, Rendeng Shi, Weilin Zhang, Dawen Zhang, Bingshuai Li
**Project Administration:** Maodu Yan

© 2022. American Geophysical Union. All Rights Reserved.

# New Paleomagnetic and Chronological Constraints on the Late Triassic Position of the Eastern Qiangtang Terrane: Implications for the Closure of the Paleo-Jinshajiang Ocean

**Liang Yu[1,2], Maodu Yan[1], Mathew Domeier[3], Chong Guan[4], Miaomiao Shen[1], Qiang Fu[1,2], Wanlong Xu[1,2], Zunbo Xu[1,2], Zhichao Niu[1,2], Liye Yang[5], Rendeng Shi[1], Weilin Zhang[1], Jinbo Zan[1], Dawen Zhang[6], and Bingshuai Li[7]**

[1]State Key Laboratory of Tibetan Plateau Earth System, Resources and Environment (TPESRE), Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing, China, [2]University of Chinese Academy of Sciences, Beijing, China, [3]Centre for Earth Evolution and Dynamics (CEED), University of Oslo, Oslo, Norway, [4]Xi'an Center of Geological Survey (Northwest China Center of Geoscience Innovation), China Geological Survey, Xi'an, China, [5]Pu'er Natural Resources and Planning Bureau, Pu'er, China, [6]College of Tourism and Environmental Resource, Zaozhuang University, Zaozhuang, China, [7]School of Earth Sciences, East China University of Technology, Nanchang, China

**Abstract** The tectonic evolution of the Paleo-Jinshajiang Ocean, and in particular the time of its closure, is debated. Here we present new constraints on the evolution of this region from an integrated paleomagnetic and geochronologic study of the Late Triassic Batang Group volcanic rocks in Qamdo, Eastern Qiangtang Terrane (EQT). Two zircon U-Pb ages indicate that the volcanic rocks erupted at ~227–222 Ma. New paleomagnetic results yield a robust Late Triassic paleopole of 57.6°N, 176.4°E ($A_{95}$ = 7.8°), corresponding to a paleolatitude of 32.5 ± 7.8°N for the study area. Integrated with other lines of paleomagnetic and geological evidence, we show that the northward drift of the EQT placed it at the same paleolatitude as the Tarim Block by ~227–222 Ma, which suggests that the Paleo-Jinshajiang Ocean in the Qamdo region closed before that time, and likely around ~230 Ma.

**Plain Language Summary** The closure of the Paleo-Jinshajiang Ocean, which was one of the most important ancient oceans of Asia, closed when the Qiangtang Terrane (QT) collided with the terranes now located to the north of it. However, the evolution of this ocean, and especially the timing of its closure, are still debated. The drift history of the QT is a key to understanding the evolution of the Paleo-Jinshajiang Ocean. We present here new constraints on the evolution of this region from an integrated paleomagnetic and zircon U-Pb geochronologic study of Late Triassic volcanic rocks in Qamdo, Eastern QT. Two zircon U-Pb geochronologic results indicate that the volcanic rocks erupted at ~227–222 Ma. New paleomagnetic results yield a robust Late Triassic paleopole of 57.6°N and 176.4°E, and indicate that the study area was then located at a paleolatitude of 32.5 ± 7.8°N. In integrating these data with other lines of evidence, we show that a northward drift of the EQT since the early Permian brought it to the same paleolatitude as the Tarim Block by ~227–222 Ma. This implies that the closure of the Paleo-Jinshajiang Ocean in the Qamdo region occurred before 227–222 Ma, and likely around ~230 Ma.

## 1. Introduction

Asia is a mosaic of continental blocks and accretionary systems amalgamated together along sutures marking lost oceans. One of the most important of these ancient oceans—the Paleo-Tethys—was an immense Paleozoic-early Mesozoic ocean which, in its western sector, once separated Gondwana and Laurasia (Sengör, 1979). In its eastern half, the Paleo-Tethys separated a collection of proto-Asian continental blocks in the north (the Asiatic Hunic terranes of Stampfli et al., 2002) from the northeast margin of Gondwana in the south (G. C. Zhao et al., 2018). A common first-order narrative envisions the Paleo-Tethys as a single, wedge-shaped basin that ultimately closed by the arrival of a ribbon-shaped string of Gondwana-derived microcontinents (the Cimmerian terranes) to the southern margin of Laurasia and proto-Asia in the Mesozoic. However, in the east, the structure of the Paleo-Tethys and its closure was probably more complex, involving at least two oceanic basins, or 'branches' of the Paleo-Tethys. A northern branch, which is often called the Paleo-Jinshajiang Ocean (or the Mianlue Ocean), separated the Tarim (TB), Qaidam, Central Qilian, Alxa (or Alex) and North China (NCB) blocks in the north from the Eastern Qiangtang Terrane (EQT) and the Indochina and South China blocks in the south. A

**Resources:** Liang Yu, Maodu Yan, Chong Guan, Miaomiao Shen, Qiang Fu, Wanlong Xu, Zunbo Xu, Zhichao Niu, Liye Yang, Dawen Zhang, Bingshuai Li
**Software:** Liang Yu, Chong Guan, Miaomiao Shen, Qiang Fu, Wanlong Xu, Zunbo Xu, Zhichao Niu, Dawen Zhang, Bingshuai Li
**Supervision:** Maodu Yan, Mathew Domeier, Rendeng Shi, Weilin Zhang, Jinbo Zan
**Validation:** Maodu Yan
**Visualization:** Liang Yu, Mathew Domeier, Wanlong Xu, Jinbo Zan, Bingshuai Li
**Writing – original draft:** Liang Yu, Maodu Yan
**Writing – review & editing:** Maodu Yan, Mathew Domeier, Chong Guan, Miaomiao Shen, Qiang Fu, Wanlong Xu, Zunbo Xu, Liye Yang, Rendeng Shi, Weilin Zhang, Jinbo Zan, Dawen Zhang, Bingshuai Li

southern branch, often called the Longmu Tso-Shuanghu Ocean, separated the EQT and Indochina-South China blocks from the northeast margin of Gondwana and the Sibumasu and Western Qiangtang-Lhasa terranes which were likely peripheral to it in late Paleozoic time (Jian et al., 2009a, 2009b; Li, 2008; Zhai et al., 2011; G. C. Zhao et al., 2018; Figure 1a). Both of these branches may have represented substantial oceans at different times (both have been argued to constitute the 'main' branch of the eastern Paleo-Tethys) and they may have opened and closed by the migration of terranes which spalled sequentially from northeastern Gondwana. Knowledge of the individual histories of these basins is thus important to understanding of the Paleo-Tethys more generally. In this study, our focus is the Paleo-Jinshajiang Ocean.

The Paleo-Jinshajiang Ocean appeared at least by mid-Paleozoic time and in the late Paleozoic and may have spanned a width of more than 5,000 km (Guan et al., 2021). This once-expansive ocean ultimately closed when the Gondwana-derived EQT collided with the TB and the NCB, but the specific timing of that event remains critically uncertain (e.g., Guan et al., 2021; Reid et al., 2005; P. P. Song et al., 2015; M. D. Yan et al., 2016; Zi et al., 2012). On the basis of a variety of approaches (e.g., paleontologic, paleomagnetic, magmatic, sedimentologic, structural), proposed closure times for the Paleo-Jinshajiang Ocean range from the late Permian (Sha & Fürsich, 1999) to the Early–Middle Triassic (Pullen et al., 2008; Zi et al., 2012), the Late Triassic (Guan et al., 2021; Li, 2008; P. P. Song et al., 2015; M. D. Yan et al., 2016), and the latest Triassic–Early Jurassic (Liu, Tan, et al., 2019; Roger et al., 2010; Tan et al., 2020; L. Y. Zhang et al., 2014). Clearly there is a need for more precise and direct constraints on the timing of this important event.

The closure of east-west oriented oceans can be quantitatively established from paleomagnetism, which can determine the latitudinal drift history (and eventual collision) of the continental blocks bordering an ocean. Given that the EQT is immediately south of the Jinshajiang Suture Zone (Figure 1a), its drift history is important to understand the evolution and closure of the Paleo-Jinshajiang Ocean. A number of Permian to Triassic paleomagnetic studies have already been carried out in the EQT in order to detail the evolution of the Paleo-Jinshajiang Ocean (e.g., Cheng et al., 2012, 2013; Guan et al., 2021; K. N. Huang et al., 1992; X. Z. Li et al., 1999; Lin & Watts, 1988; Ma et al., 2019; C. Y. Song et al., 2012; P. P. Song et al., 2015, 2017, 2020; X. F. Yang et al., 2017; Zhou et al., 2019; see Figure S1 in Supporting Information S1 for locations). However, collectively, these data only constrain the closure time of the Paleo-Jinshajiang Ocean to between ~240 and 209 Ma (Guan et al., 2021), and so again a more precise estimate of its timing of closure is needed.

In this study, we report new joint paleomagnetic and U-Pb geochronologic data from Late Triassic volcanic rocks of the EQT. Based on these new data and a synthesis of reported paleomagnetic data and other relevant geological constraints, we further describe the drift history of the EQT during the Permian and Triassic and more precisely estimate the closure time of the Paleo-Jinshajiang Ocean in this region.

## 2. Geological Setting and Sampling

The EQT (alternatively called the 'North Qiangtang Terrane' in the literature) comprises one half of the larger, elongate Qiangtang Terrane, which varies in width from ~500 km in its mid-section to ~150 km in its eastern and western ends (Figure 1a). To the south, the EQT is separated from the Western Qiangtang Terrane (WQT; alternatively called the 'South Qiangtang Terrane') by the so-called Longmu Tso-Shuanghu Suture Zone, and to the north and northeast the Jinshajiang Suture Zone separates the EQT from the Songpan-Ganzi Terrane and the Yidun Terrane, respectively (e.g., BGMRXAR, 1993; C. Li et al., 2009; Y. Liu et al., 2011; Metcalfe, 2013; Pan et al., 2004; Zhang, Cai, et al., 2006; Zhang, Zhang et al., 2006; Zhu et al., 2013; Figure 1a). Note that the east-west continuity of the Jinshajiang Suture Zone across the length of the EQT is a matter of some debate, particularly east and west of the Zhiduo area. It is also worth noting that despite bearing the same 'Qiangtang' label, the EQT and WQT were probably independent of each other in late Paleozoic time: the WQT hosts Permo–Carboniferous glaciomarine diamictites and cold-water biota of Gondwana-affinity whereas the EQT hosts Permo–Carboniferous carbonates with warmer-water faunas of Cathyasian affinity (Y. C. Zhang et al., 2013). It is presently unknown what comprises the basement of the EQT, and the oldest exposed rocks in the terrane are meta-sedimentary rocks of the Early Ordovician Ningduo Group. The Ningduo Group is unconformably overlain by Late Devonian to late Permian strata, but the most widespread sedimentary sequences in the EQT are of Mesozoic and Cenozoic age (Z. Z. Zhao et al., 2001).



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2021GL096902

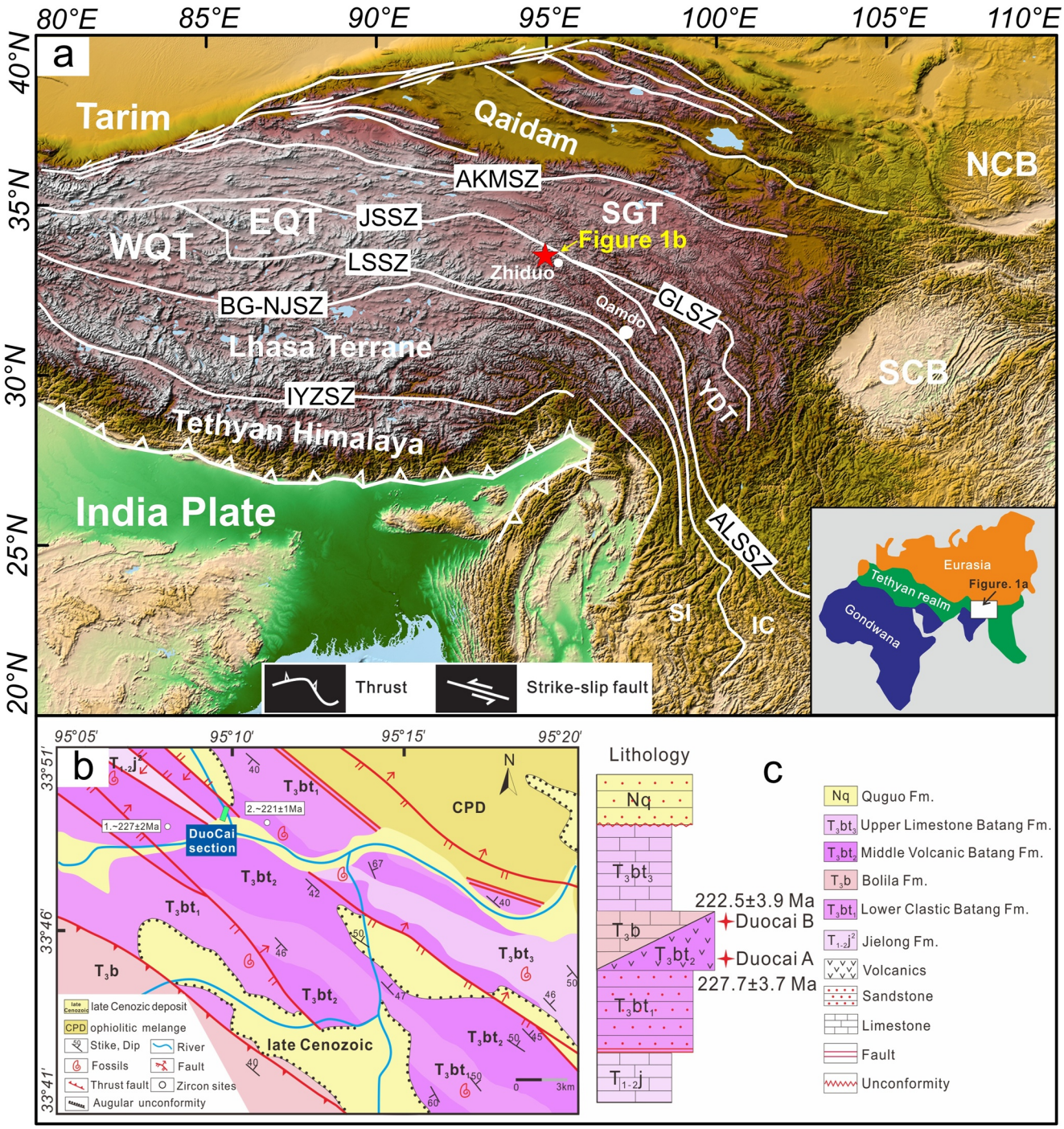



**Figure 1.** (a) Topographic map of the Tibetan Plateau and surrounding areas with terranes and major faults, with the inset map showing the Tethys tectonic realm. The red star marks the location of the study area, illustrated in (b). Note the Jinshajiang Suture Zone (JSSZ) is here taken to represent the suture of the Paleo-Jinshajiang Ocean, whereas the Longmu Tso-Shuanghu Suture Zone (LSSZ) represents the suture of the Longmu Tso-Shuanghu Ocean. (b) Detailed geologic map of the Zhiduo area showing the location of the sampled Duocai section in green square. (c) Lithologic sequence of the Triassic and late Cenozoic strata in the Zhiduo area (modified from QGSI, 2005a) in the Eastern Qiangtang Terrane. Abbreviations in (a): ALSSZ, Ailaoshan Suture Zone; AKMSZ, Ayimaqing-Kunlun-Muztagh Suture Zone; BG-NJSZ, Bangong-Nujiang Suture Zone; EQT, Eastern Qiangtang Terrane; GLSZ, Ganzi-Litang Suture Zone; IC, Indochina Terrane; IYZSZ, Indus-Yarlung Zangbo Suture Zone; NCB, North China Block; SCB, South China Block; SGT, Songpan-Ganzi Terrane; SI, Sibumasu Terrane; WQT, Western Qiangtang Terrane; YDT, Yidun Terrane. The white circles with ages in (b) represent the U-Pb zircons sites of previous studies: (1) Liu, Tan, et al. (2019); (2) S. Q. Zhao et al. (2014). The red stars in (c) represent the geochronologic sampling positions in the Middle Volcanic Batang Formation.



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Our study is focused on the Zhiduo area, which is located in the Qamdo region that is adjacent to the Jinshajiang Suture Zone in the east of the EQT (Figure 1a). In this area, the principal lithologic units comprise the Carboniferous to early-middle Permian Duocai ophiolitic mélange (sandstone, slate and volcanic rock), the Early–Middle Triassic Jielong Formation (breccia and bioclastic limestone), the Late Triassic Bolila Formation (biolimestone and limestone breccia), and the Late Triassic Batang Group, above which late Cenozoic deposits locally lie unconformably (the Neogene Quguo Formation and Quaternary deposits; QGSI, 2005a; X. M. Zhao et al., 2006; Figures 1b and 1c). The Batang Group is widely distributed in the Zhiduo area, which is bounded by the Duocai ophiolitic mélange to the north and the Bolila Formation to the south. The Batang Group can be further divided into three formations: the Lower Clastic, the Middle Volcanic and the Upper Limestone Batang formations (QGSI, 2014; X. M. Zhao et al., 2006). The Middle Volcanic Batang Formation is considered to be a temporal correlative of the Bolila Formation limestone (QGSI, 2005a; Figure 1c). Abundant fossils, including brachiopods (e.g., *Rhaetinopsis pentagonalis Ching*, *Oxycolpella elongata Ching et Fang*, *Amphiclina* sp., *Arcestes* sp.), bivalves (*Pteria angusta* (*Saurin*), *Neomegalodon* (*N.*) sp.), corals (*Montlivaltia norica Frech*, *Thecosmilia sinensis Liao et Li*, *Pinacophyllum* sp.) and ammonites (*Paratibetites* sp.), have been observed within the Lower Clastic and Upper Limestone Batang formations in the Zhiduo area. These fossils constrain the age of the Batang Group to the Late Triassic (Carnian–Norian) and indicate its deposition occurred in a shallow marine environment (QGSI, 2014; X. M. Zhao et al., 2006).

The Middle Volcanic Batang Formation is well exposed in the Zhiduo area (Figure 1b), due to regional NW-trending faults. The exposure of the formation extends more than 10 km laterally, and it presents a thickness of up to 1,000 m. It consists mainly of basalt, rhyolitic tuff, rhyolite, dacite, and crystal tuff, and is intercalated with some limestone and siltstone layers (QGSI, 2014; X. M. Zhao et al., 2006). The Middle Volcanic Batang Formation has conformable contacts with both the underlying Lower Clastic and overlying Upper Limestone Batang formations. Two recent U-Pb zircon dating results from dacites and rhyolites of the Middle Volcanic Batang Formation in the Zhiduo area have yielded crystallization ages of 227 ± 2 Ma (Liu, Tan, et al., 2019) and 221 ± 1 Ma (S. Q. Zhao et al., 2014; Figure 1b). Given the clear bedding of the formation and the fact that it was emplaced within the critical age range that we are interested in (~240–209 Ma), we elected to target the thick Middle Volcanic Batang Formation for paleomagnetic study.

We collected samples from the Middle Volcanic Batang Formation at the Duocai section (33.8°N, 95.2°E) along the Riari oblique syncline in Zhiduo. In this section, the volcanic rocks mostly comprise lithic-crystal tuff and crystal tuff (Figures S2e and S2f in Supporting Information S1). The syncline has a NW-SE-trending axis with a northeast limb dipping ~25°–53° toward the southwest and a southwest limb dipping ~52° toward the northeast. Given the absence of Jurassic-Cretaceous sediments in the Zhiduo area, the exact time of folding cannot be constrained. However, in the adjacent Zaduo area, an angular unconformity is observed between Late Triassic strata and the Middle Jurassic Quemocuo Formation (QGSI, 2014), which implies that a major phase of deformation occurred in the region during the Late Triassic or Early Jurassic. We collected paleomagnetic samples from both fold limbs in the Duocai section, which is over 200 m thick. Eight to twelve core samples were collected at each site with the use of a portable gasoline-powered drill. The paleomagnetic samples were oriented with a magnetic compass and also a sun compass when weather allowed. A total of 131 core samples of tuff were collected from 13 sampling sites (sites 1–4 and 8–11 come from the southwest limb and sites 5–7, 12 and 13 come from northeast limb; Figures S2a–S2d and S2g in Supporting Information S1). In addition, samples of crystal tuff and lithic-crystal tuff were collected for U-Pb geochronology from the stratigraphic lower (Duocai A) and upper parts (Duocai B) of the section (Figure 1c and Figure S2g in Supporting Information S1).

## 3. Zircon U–Pb Geochronology

Zircons were extracted from the two bulk samples using routine crushing and heavy mineral separation techniques at the Yu-Neng Technology Service Limited Company for rock and mineral separation, Langfang, China. Transmitted and reflected light photomicrographs and cathodoluminescence (CL) images were obtained to analyze the internal textures of the zircons and to select target sites for U-Pb dating. Zircon U-Pb geochronologic analyses were conducted at the State Key Laboratory of Tibetan Plateau Earth System, Resources and Environment, Institute of Tibetan Plateau Research, Chinese Academy of Sciences in Beijing, China, using laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS). In this study, zircon standards 91500 and GJ-1 were used as our primary and secondary standards to determine the U-Th-Pb ratios, and NIST SRM 610 was used

19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

as an external standard to calibrate the Pb concentrations. Isotope ratios and concentrations were obtained by using the GLITTER 4.0 software (Griffin et al., 2008). In this study, we apply a <10% discordance filter to the generated data. U-Pb concordia diagrams and weighted average plots were made using Isoplot 4.0 (Ludwig, 2012).

The CL images show that the zircon grains from both samples (Duocai A and Duocai B) are generally transparent, euhedral or subhedral, and have a prismatic shape with clear oscillatory zoning (Figures S3a and S3d in Supporting Information S1). The lengths of the long axes of the grains are 60–150 μm and their Th/U ratios are between ∼0.3 and 0.8, consistent with a magmatic origin (Table S1 in the Supporting Information S1). Analyses of 23 and 19 zircon grains yield a weighted mean $^{206}Pb/^{238}U$ age of 227.7 ± 3.7 Ma (2σ; MSWD = 0.19) for the stratigraphic lower sample (Duocai A) and 222.5 ± 3.9 Ma (2σ; MSWD = 0.117) for the stratigraphic upper sample (Duocai B; Figure S3 in Supporting Information S1), respectively.

These two U-Pb zircon ages from the lower and upper parts of the Duocai section suggest that the sampled Middle Volcanic Batang Formation ranges in age at least from ∼227 to 222 Ma. This age range is consistent with both the Carnian–Norian biostratigraphic age estimate on the Batang Group in the Zhiduo region (QGSI, 2014; X. M. Zhao et al., 2006), and with the two existing U-Pb zircon ages of 227 ± 2 Ma (Liu, Tan, et al., 2019) and 221 ± 1 Ma (S. Q. Zhao et al., 2014; Figure 1b) from the Middle Volcanic Batang Formation in the vicinity of the study area.

## 4. Paleomagnetic Results

Rock magnetic experiments and paleomagnetic treatments are described in Text S1 and S2 in the Supporting Information S1. In summary, these rock magnetic results suggest that SD–PSD magnetite is the main magnetic carrier (in a few cases also with hematite) in the Middle Volcanic Batang Formation in the Duocai section (Figure S4 in Supporting Information S1). A total of 86 out of 132 samples yielded stable ChRM directions that fulfill our quality criteria (Text S2 in Supporting Information S1), allowing the definition of a site mean from 11 out of the 13 sites collected. Of these 11 sites, 7 exhibit dual-polarity directions, whereas the other 4 have only normal polarity directions. Together, they provide an overall mean direction of $D_g = 62.5°$, $I_g = 58.7°$, $k_g = 2.6$, $\alpha_{95} = 36.2°$ *in-situ*, and $D_s = 38.9°$, $I_s = 51.9°$, $k_s = 45.3$, and $\alpha_{95} = 6.9°$ after tilt correction (Figure 2c, Tables S3 and S4 in Supporting Information S1). With tilt-correction, these data yield positive McFadden (1990) and Watson and Enkin (1993) foldtests, a positive McFadden and McElhinny (1990) reversal test (see details in Text S3 of Supporting Information S1), and pass a paleosecular variation check (Text S3 in Supporting Information S1). Collectively, these results suggest that the ChRM likely represents a primary magnetization. These data yield a paleopole of 57.6°N, 176.4°E, $A_{95} = 7.8°$, which corresponds to a paleolatitude of 32.5° ± 7.8°N for the study area (Table S4 in Supporting Information S1).

## 5. Discussion

### 5.1. Paleomagnetic Constraints on the Permian-Triassic Drift of the EQT

To date, a total of 18 Permian to Triassic paleomagnetic results have been reported from the EQT (Cheng et al., 2012, 2013; Guan et al., 2021; K. N. Huang et al., 1992; X. Z. Li et al., 1999; Lin & Watts, 1988; Ma et al., 2019; C. Y. Song et al., 2012; P. P. Song et al., 2015, 2017, 2020; X. F. Yang et al., 2017; Zhou et al., 2019) which present quantitative constraints on its drift history. However, these data are of variable quality and a detailed evaluation by Guan et al. (2021) determined that only five of the poles are of sufficiently high quality to be of use in reconstructing the drift history of the EQT (Table S5 in Supporting Information S1). According to these constraints, the EQT was located in the low latitudes of the Southern Hemisphere in the early Permian (∼22°S at ∼297 Ma; P. P. Song et al., 2017), but drifted rapidly northward to cross the equator in the latest Permian (∼5°S at ∼259 Ma and ∼11°N at ∼251 Ma; Guan et al., 2021; Ma et al., 2019), and reached ∼25°N by ∼240 Ma (P. P. Song et al., 2020) and then reached ∼35°N by the latest Triassic (at ∼209 Ma; P. P. Song et al., 2015; Figures 3a and 3b).

In Figure 3b, we compare this latitudinal drift history of the EQT against paleolatitude curves for Eurasia, the Tarim Block (TB) and India (Gondwana), as computed for the same reference location (Table S6 in Supporting Information S1). As can be seen from these comparisons, the EQT was located some 3,000 km north of the Indian margin of Gondwana in the earliest Permian, which is consistent with the observation that the EQT hosts

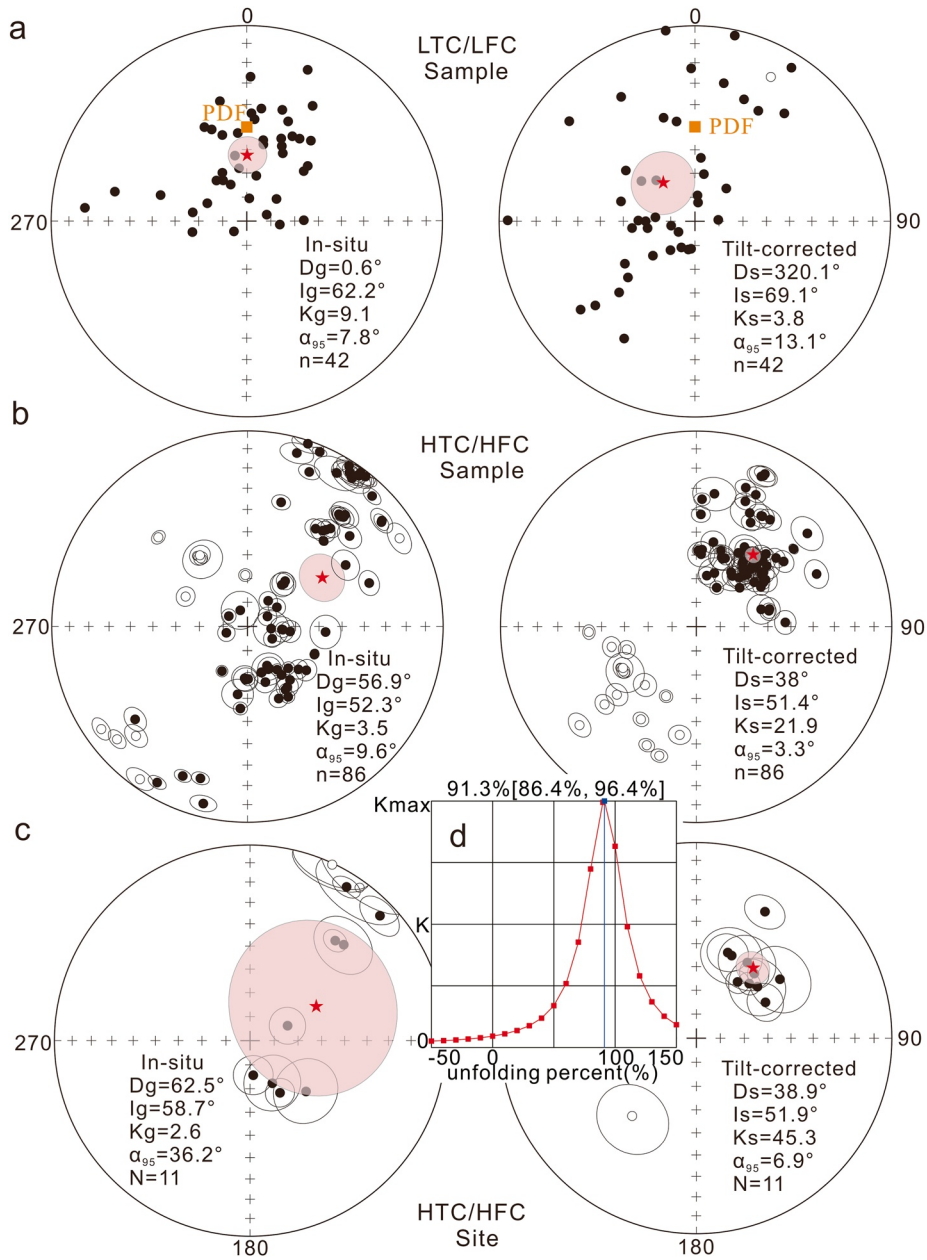



**Figure 2.** Equal-area projections of the viscous overprint (low-temperature (field) component) directions (a), characteristic remnant magnetization directions of specimens (b) and site mean directions (c) for the Middle Volcanic Batang Formation, and the progressive unfolding analysis of the ChRM directions using the Watson and Enkin (1993) method (d). PDF = direction of the present-day geomagnetic field in (a). Open/solid circles represent upward/downward inclinations in (a), (b) and (c).

warm-water Cathyasian faunas of this age (in contrast to the WQT, which hosts cold-water Cimmerian/Gondwanan biotas of Permo–Carboniferous age). At the same time, the EQT was also then separated from Eurasia and the TB by a wide Paleo-Jinshajiang Ocean that latitudinally spanned at least 5,000 km (Figure 3b). However, once the EQT began to drift rapidly northward from mid-Permian time, the Paleo-Jinshajiang Ocean began to shrink, and by the Middle Triassic the EQT was only ~1,000 km from margin of the TB. It could be anticipated that the collision between the EQT and the TB followed shortly thereafter, and Guan et al. (2021) inferred their collision to have occurred at ~230 Ma by extrapolating the northward drift of the EQT into the Late Triassic. However, among these existing data, the next oldest direct paleomagnetic constraint on the EQT, which yields a



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

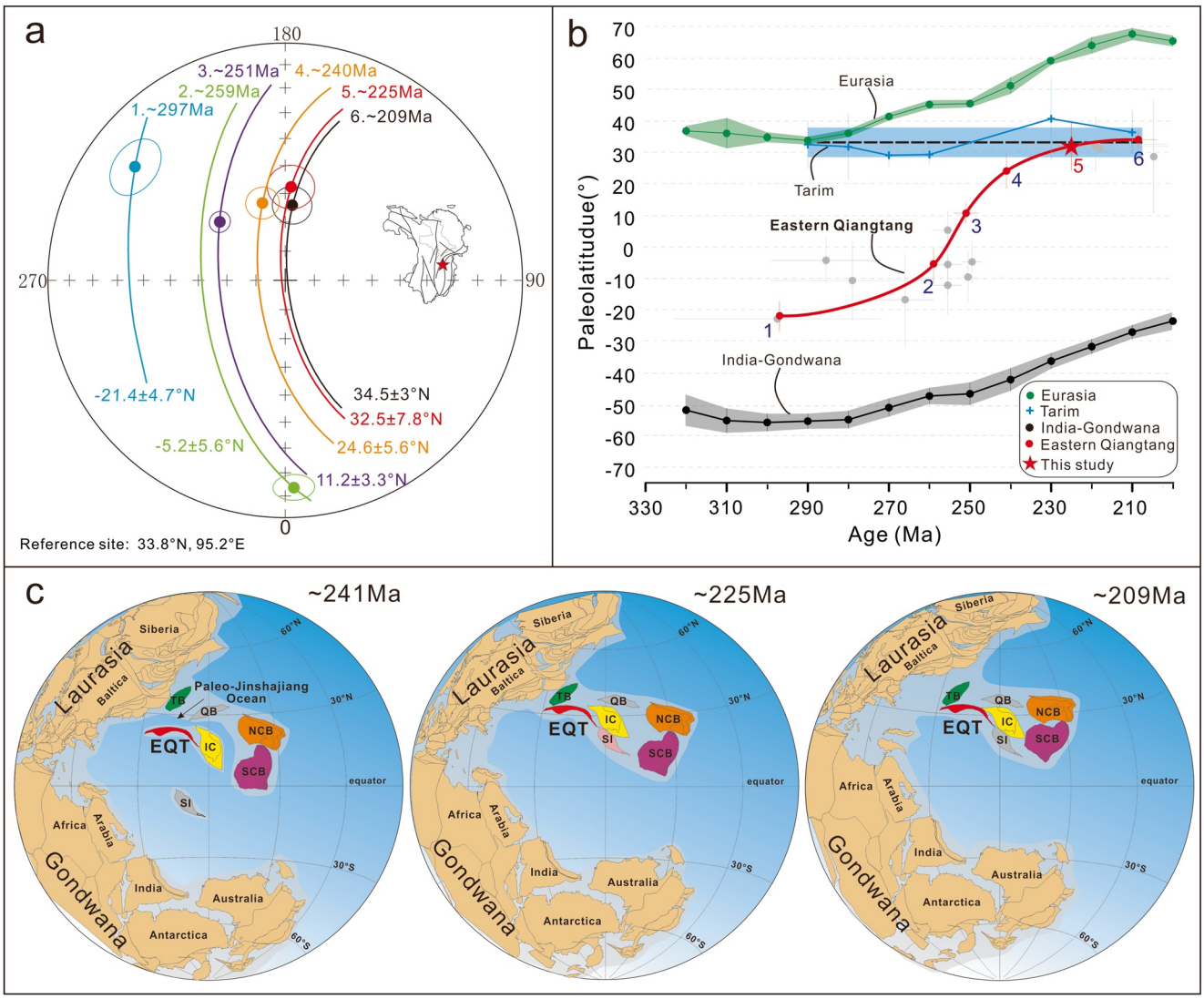



**Figure 3.** Equal-area projection of the reliable paleomagnetic poles of the EQT (reference point: 33.8°N, 95.2°E) (a), paleolatitude evolution of the EQT, TB, Eurasia and Gondwana (b) during ~297–209 Ma (modified from Guan et al., 2021) and paleogeographic reconstructions of the EQT at 241 Ma, 225 Ma and 209 Ma (c). Small-circle fitting (colored line) passing through paleopoles (colored circle) and centered on the reference point (red star) in (a). Paleolatitudes of Eurasia and India-Gondwana are from Torsvik et al. (2012); those of the TB are from B. C. Huang et al. (2018) (Table S6 in Supporting Information S1) in (b). Paleomagnetic poles of the EQT from: 1, P. P. Song et al. (2017); 2, Ma et al. (2019); 3, Guan et al. (2021); 4, P. P. Song et al. (2020); 5, This study; 6, P. P. Song et al. (2015) in (a) and (b) (Table S5 in Supporting Information S1). Abbreviations in (c): EQT, Eastern Qiangtang Terrane; IC, Indochina Terrane; NCB, North China Block; QB, Qaidam Block. SCB, South China Block; SI, Sibumasu Terrane; TB, Tarim Block. The EQT was located at a paleolatitude of 24.6 ± 5.6°N at ~241 Ma, implying there existed a ~1,000 km wide Paleo-Jinshajiang Ocean between the EQT and the TB then. By ~225 Ma the EQT at ~32.5° ± 7.8°N shows an indistinguishable paleolatitude with that of the TB. Similar paleolatitudes are observed for EQT, IC and SI at ~209 Ma. The reconstructions were performed with GMAP (Torsvik et al., 2012). Colored Asian tectonic blocks are paleomagnetically constrained according to the following: EQT: P. P. Song et al. (2020), IC: Yan, Zhao, Zhang, Huang, et al. (2019) at ~241 Ma; EQT: This study, IC: Y. G. Yan et al. (2017), SI: J. Zhao et al. (2015) at ~225 Ma; EQT: P. P. Song et al. (2015), IC: Yan, Zhao, Zhang, Charusiri, et al. (2019) at ~209 Ma; NCB, SCB, TB: B. C. Huang et al. (2018) in (c). Gray Asian terranes (blocks) are those with no paleomagnetic constraints. Paleolatitudes were determined based on Tables S5 and S6 in Supporting Information S1. Note that the Western Qiangtang Terrane is not shown in the figure, due to a lack of paleomagnetic data and geological constraints on its positioning.

paleolatitude indistinguishable from that of the TB, is latest Triassic in age (~213–204 Ma (mean of 209 Ma); Figures 3a and 3b; pole #6). Thus, while an early Late Triassic collision between the EQT and the TB seems probable, these paleomagnetic data only constrain that collision between ~240 and 209 Ma.

Our new paleomagnetic result from the Late Triassic Middle Volcanic Batang Formation provides a direct constraint on the paleolatitude of the EQT during this critical interval. The constituent data yield both positive fold



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and reversals tests and the Late Triassic host rocks are well-dated. The new paleomagnetic pole (57.6°N, 176.4°E, $A_{95}$ = 7.8°) fulfills all seven of the reliability criteria of Van der Voo (1990) and Meert et al. (2020) and thus can provide a robust estimate of the position of the EQT at ~227–222 Ma (mean of ~225 Ma). At the reference locality (33.8°N, 95.2°E), this result corresponds to a paleolatitude of ~32.5 ± 7.8°N (Figures 3a and 3b; pole #5). This paleolatitude is indistinguishable from the average Permian–Triassic paleolatitude of the TB (~33.8° ± 4.4°N, B. C. Huang et al., 2018; Table S6 in Supporting Information S1), suggesting that closure of the Paleo-Jinshajiang Ocean by collision of the EQT and the TB/Songpan-Ganzi/Yidun Terrane had occurred at least by (but possibly prior to) 227–222 Ma (Figure 3). This constraint is thus consistent with a ~230 Ma collision, as inferred by Guan et al. (2021) from their extrapolation of a constant northward drift of the EQT since the Permian.

### 5.2. Insights Into the Closure Time of the Paleo-Jinshajiang Ocean

While our new paleomagnetic result presents a key constraint on the closure time of the Paleo-Jinshajiang Ocean, we note some important caveats associated with this simple analysis. First, the Jinshajiang Suture Zone marks the divide between the EQT (to the south) and Songpang-Ganzi and Yidun terranes (to the north and northeast), but there are no reliable Triassic paleomagnetic data available from the latter terranes and so we can only compare results from the EQT and the TB (which do not directly share a suture). Second, a shallow inclination bias is commonly observed among Cretaceous–Cenozoic sedimentary rocks of Central Asia (including the TB) and the Triassic paleomagnetic data from the TB are mostly derived from clastic sedimentary rocks (B. C. Huang et al., 2018), so the possibility of a shallow inclination bias cannot be ruled out. If the Triassic data from the TB are contaminated by such bias, the collision of the EQT and the TB could have occurred later than ~230 Ma. Notably, B. C. Huang et al. (2018) argued that such bias was not likely significant given the occurrence of similar paleomagnetic directions in Jurassic sedimentary and igneous rocks from the TB (but see also Wu et al., 2017), and in the absence of demonstrated bias, we have here used their compilation, unaltered. Nevertheless, future work on the acquisition of new high-quality Triassic paleomagnetic data from the TB and the terranes between it and the EQT are clearly needed to provide more robust constraints.

Geological estimates of the closure time of the Paleo-Jinshajiang Ocean vary in the literature from at least the late Permian to the Jurassic. This range can be attributed, at least in part, to the use of different methods which track different processes (e.g., paleontologic, magmatic, geochemical, sedimentologic, structural), but also to disagreements about the location and lateral continuation of the Paleo-Jinshajiang Ocean. Nevertheless, along the Jinshajiang Suture Zone, which we are concerned with herein, there are a number of observations that are consistent with a Late Triassic closure of the Paleo-Jinshajiang Ocean.

The maximum closure time of an ocean can be determined from the age of the youngest ophiolitic material preserved within the suture, which must pre-date final suturing. In the Zhiduo area, mafic rocks as young as ~234 ± 3 Ma (U-Pb on zircon) have been reported from the ophiolitic mélange in the suture zone (B. Liu et al., 2016) and the occurrence of ocean island basalts with ages spanning from ~422 to 236 Ma (U-Pb on zircon) confirms that the suture represents a major, long-lived ocean (Liu, Xiao, et al., 2019) prior to ~234 Ma. To the east of the Zhiduo area, ophiolitic material in the Jinshajiang Suture Zone is mostly of Carboniferous age, but ages as young as ~294 Ma have been reported (Wang et al., 2000; Zi et al., 2012), whereas to the west of the Zhiduo area, the youngest ages are Triassic: ~246 ± 5.18 Ma (K-Ar on diabase; Y. Li et al., 2003) and ~232 ± 11 Ma (Sm-Nd isochron from metagabbro; N. Zhang et al., 2012). Collectively, these observations suggest that the Paleo-Jinshajiang Ocean remained open through the late Paleozoic and into the beginning of the Triassic.

The timing of ocean closure can also be constrained from structural and sedimentologic evidence of the continent-continent collision concluding it. Along the Jinshajiang Suture Zone in the Qamdo region, prominent unconformities have been reported between the Middle Triassic Jielong Formation and the Late Triassic Jieza Group in the Zaduo and Zhiduo area (QGSI, 2005a, 2005b), and between the Early−Middle Triassic Malasongduo Formation and the Late Triassic Jiapila Formation in the Markam area (TARGS, 2007). In the suture itself in the Yushu area, ophiolitic rocks including a Late Triassic mélange are thrusted over a Late Triassic matrix (Dewey et al., 1988; Metcalfe, 1996) and Late Triassic structural fabrics have been dated to ~230.5 ± 0.8 Ma by Ar-Ar on phengite (T. N. Yang et al., 2012). In the Gonjo area to the southeast, Middle Triassic eclogites (~240 ± 3 Ma and ~244 ± 1 Ma; U-Pb on zircon) have been discovered in the suture zone, some of which are hosted by a ~226 Ma (U-Pb on zircon) adamellite, implying that they were rapidly exhumed in latest Middle to early Late Triassic time (Tang et al., 2020). To the north, in the Songpan-Ganzi Terrane, the Late Triassic was

19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

likewise marked by deformation (~230–200 Ma; Harrowfield & Wilson, 2005) and the initiation of non-marine deposition. Other sedimentologic evidence of a Late Triassic collision include the appearance of Late Triassic (Rhaetian palynological age) molasse unconformably above Early–Middle Triassic rocks along the northern margin of the EQT—which has been attributed to the development of a foreland basin associated with collision of the EQT against Songpan-Ganzi Terrane (BGMRXAR, 1993; Y. Li et al., 2003)—and the initiation of sediment mixing between those terranes already by Carnian time (Ding et al., 2013; Gong et al., 2021). Looking further afield, G. C. Zhao et al. (2018) recently presented a regional geologic synthesis arguing that the East Kunlun (to the north of the Jinshajiang Suture Zone) and the Mianlue-Qinling and Dabie-Sulu areas (to the east) all signify closure of the regional Mianlue (e.g., Paleo-Jinshajiang) Ocean at ~240–220 Ma. Together, these observations are consistent with a Late Triassic closure of the Paleo-Jinshajiang Ocean.

Many arguments for a younger (Early Jurassic) closure of the Paleo-Jinshajiang Ocean are predicated on interpretations of the magmatic record. The closure of an ocean necessitates the consumption of oceanic lithosphere by subduction, and so the timing of arc-related magmatism and its replacement by syn- to post-collisional magmatism can be used to constrain the age of closure. Unfortunately, while there are an abundance of Triassic igneous rocks in the northern EQT and southern Songpan-Ganzi Terrane, the tectonic environment of their emplacement—as mostly inferred from geochemistry—remains a topic of active debate. For example, in the eastern Songpan-Ganzi Terrane, H. F. Zhang et al. (2007) have interpreted ~211 Ma A-type granites to be post-collisional, whereas Yuan et al. (2010) interpreted ~224–205 Ma adakitic and shoshonitic plutons in the eastern Songpan-Ganzi Terrane to result from crustal thickening during collision, and L. Y. Zhang et al. (2014) related ~225–193 Ma I-type granitoids in the western Songpan-Ganzi Terrane to ongoing subduction. Although it is beyond the scope of the present study to dissect the various geochemical arguments presented for these rocks as well as the Triassic igneous rocks along the north of the EQT, we note that our new paleomagnetic result can contribute to this discussion by placing these magmatic episodes in a more well-defined paleogeographic context.

In summary, a range of geological evidence (see Figure S1 in Supporting Information S1 for locations, Table S7 in Supporting Information S1) from along the Jinshajiang Suture Zone and our new paleomagnetic results (together with the existing ones) are consistent with a Late Triassic (~230 Ma) closure of the Paleo-Jinshajiang Ocean. On the other hand, the paleolatitudinal histories of the TB, Songpan-Ganzi Terrane and Yidun Terrane remain poor to unconstrained in early Mesozoic time, and so important uncertainties remain about this specific timing and the potential diachroneity of Paleo-Jinshajiang Ocean. Future research dedicated to acquiring new paleomagnetic constraints from these blocks to supplement the now well-resolved Permian and Triassic paleolatitudinal history of the EQT will be decisive in addressing these outstanding questions.

## 6. Conclusions

Zircon U-Pb dating of two volcanic samples from the upper and lower Middle Volcanic Batang Formation demonstrates that these rocks were erupted in the Late Triassic, between ~227 and 222 Ma. A paleomagnetic investigation of these rocks, involving detailed rock magnetism, field stability tests and paleosecular variation analysis, suggests that they maintain a primary Late Triassic magnetization. On the basis of paleomagnetic results from 11 sites (86 samples), we report a robust ~225 Ma paleomagnetic pole of 57.6°N and 176.4°E ($A_{95}$ = 7.8°). This result indicates that the EQT occupied a paleolatitude of 32.5 ± 7.8°N (reference point at 33.8°N and 95.2°E) in the early Late Triassic. Integrated with other reliable paleomagnetic data from the region, and in the context of independent geologic evidence, we ascertain that the EQT collided with the TB/Songpan-Ganzi/Yidun Terrane, and thereby closed the Paleo-Jinshajiang Ocean in the Qamdo region, before 227–222 Ma, and most likely at ~230 Ma.

## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Data Availability Statement

All data used in this study (Tables S1–S7 in Supporting Information S1) can be obtained from the fig share website (available from http://dx.doi.org/10.6084/m9.figshare.16926901).

**Acknowledgments**
This research was supported by the Second Tibetan Plateau Scientific Expedition Program (Grant 2019QZKK0707), Strategic Priority Research Program of Chinese Academy of Sciences (XDA20070201), the 2020 outstanding members of Youth Innovation Promotion Association, CAS, China (Y202023), the Natural Science Foundation of China (Grants 41988101-01, 41974080, 42164005), and the State Scholarship Fund, CSC, China (202004910721). Ziqiang Mao, Peiping Song, Zhantao Feng, Deng Zeng, Yongmin Liu, Kang Wu and Shengsheng Chen are thanked for their assistance with the laboratory work and data processing.

## References

Bureau of Geology Mineral Resources of Xizang Autonomous Region (BGMRXAR). (1993). *Regional geology of Xizang (Tibet) autonomous region* (p. 515). Geological Publishing House. (In Chinese).

Cheng, X., Wu, H. N., Diao, Z. B., Wang, H. J., Ma, L., Zhang, X. D., et al. (2013). Paleomagnetic data from the Late Carboniferous-Late Permian rocks in eastern Tibet and their implications for tectonic evolution of the northern Qiangtang-Qamdo block. *Science China Earth Sciences*, *56*(7), 1209–1220. https://doi.org/10.1007/s11430-012-4558-1

Cheng, X., Wu, H. N., Guo, Q., Hou, B. N., Xia, L. Y., Wang, H. J., et al. (2012). Paleomagnetic results of Late Paleozoic rocks from northern Qiangtang Block in Qinghai-Tibet Plateau, China. *Science China Earth Sciences*, *55*(1), 67–75. https://doi.org/10.1007/s11430-011-4287-x

Dewey, J. F., Shackleton, R. M., Chengfa, C., & Yiyin, S. (1988). The tectonic evolution of the Tibetan Plateau. *Philosophical transactions of the Royal Society of London a mathematical physical & engineering sciences*, *327*, 379–413.

Ding, L., Yang, D., Cai, F. L., Pullen, A., Kapp, P., Gehrels, G. E., et al. (2013). Provenance analysis of the Mesozoic Hoh-Xil-Songpan-Ganzi turbidites in northern Tibet: Implications for the tectonic evolution of the eastern Paleo-Tethys Ocean. *Tectonics*, *32*(1), 34–48. https://doi.org/10.1002/tect.20013

Gong, D. X., Wu, C. H., Zou, H., Zhou, X., Zhou, Y., Tan, H. Q., et al. (2021). Provenance analysis of Late Triassic turbidites in the eastern Songpan–Ganzi Flysch Complex: Sedimentary record of tectonic evolution of the eastern Paleo-Tethys Ocean. *Marine and Petroleum Geology*, *126*, 104927. https://doi.org/10.1016/j.marpetgeo.2021.104927

Griffin, W. L., Powell, W. J., & Pearson, N. J. (2008). GLITTER: Data reduction software for laser ablation ICP-MS. In P. Sylvester (Ed.), *Laser Ablation-ICP-MS in the Earth Sciences: Mineralogical Association of Canada Short Course Series* (Vol. 40, pp. 204–207).

Guan, C., Yan, M. D., Zhang, W. L., Zhang, D. W., Fu, Q., Yu, L., et al. (2021). Paleomagnetic and chronologic data bearing on the Permian/Triassic boundary position of Qamdo in the Eastern Qiantang Terrane: Implications for the closure of the Paleo-Tethys. *Geophysical Research Letters*, *48*. https://doi.org/10.1029/2020gl092059

Harrowfield, M. J., & Wilson, C. J. (2005). Indosinian deformation of the Songpan Garze fold belt, northeast Tibetan Plateau. *Journal of Structural Geology*, *27*(1), 101–117. https://doi.org/10.1016/j.jsg.2004.06.010

Huang, B. C., Yan, Y. G., Piper, J. D., Zhang, D. H., Yi, Z. Y., & Yu, S. (2018). Paleomagnetic constraints on the paleogeography of the East Asian blocks during Late Paleozoic and Early Mesozoic times. *Earth-Science Reviews*, *186*, 8–36. https://doi.org/10.1016/j.earscirev.2018.02.004

Huang, K. N., Opdyke, N. D., Peng, X. J., & Li, J. G. (1992). Paleomagnetic results from the Upper Permian of the eastern Qiangtang terrane of Tibet and their tectonic implications. *Earth and Planetary Science Letters*, *111*(1), 1–10. https://doi.org/10.1016/0012-821x(92)90164-q

Jian, P., Liu, D. Y., Kröner, A., Zhang, Q., Wang, Y. Z., Sun, X. M., et al. (2009a). Devonian to Permian plate tectonic cycle of the Paleo-Tethys Orogen in southwest China (I): Geochemistry of ophiolites, arc/back-arc assemblages and within-plate igneous rocks. *Lithos*, *113*, 748–766. https://doi.org/10.1016/j.lithos.2009.04.004

Jian, P., Liu, D. Y., Kröner, A., Zhang, Q., Wang, Y. Z., Sun, X. M., et al. (2009b). Devonian to Permian plate tectonic cycle of the Paleo-Tethys Orogen in southwest China (II): Insights from zircon ages of ophiolites, arc/back-arc assemblages and within-plate igneous rocks and generation of the Emeishan CFB province. *Lithos*, *113*, 767–784. https://doi.org/10.1016/j.lithos.2009.04.006

Li, C. (2008). A review on 20 Years' Study of the Longmu Co—Shuanghu—Lancang River Suture Zone in Qinghai—Xizang (Tibet) Plateau. *Geological Review*, *54*, 105–119.

Li, C., Zhai, G. Y., Wang, L. Q., Yin, F. G., & Mao, X. C. (2009). An important window for understanding the Qinghai-Tibet Plateau: A review on research progress in recent years of Qiangtang area, Tibet. *Geological Bulletin of China*, *28*, 1169–1177.

Li, X. Z., Liu, W. J., Wang, Y. Z., Zhu, Q. W., Du, D. X., Shen, G. F., et al. (1999). *The tectonic evolution and metallogenesis in the Tethys of the Nujiang-Lancangjiang-Jinshajiang area* (pp. 96–102). Geological Publishing House. (In Chinese).

Li, Y., Wang, C. S., & Yi, H. S. (2003). The Late Triassic collision and sedimentary responses at western segment of Jinshajiang Suture, Tibet. *Acta Sedmentologica Sinica*, *21*(2), 191–197.

Lin, J. L., & Watts, D. R. (1988). Palaeomagnetic results from the Tibetan Plateau. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *327*, 239–262. https://doi.org/10.1098/rsta.1988.0128

Liu, B., Ma, C. Q., Guo, Y. H., Xiong, F. H., Guo, P., & Zhang, X. (2016). Petrogenesis and tectonic implications of Triassic mafic complexes with MORB/OIB affinities from the western Garzê-Litang ophiolitic mélange, central Tibetan Plateau. *Lithos*, *260*, 253–267. https://doi.org/10.1016/j.lithos.2016.06.009

Liu, Y., Santosh, M., Zhao, Z. B., Niu, W. C., & Wang, G. H. (2011). Evidence for palaeo-tethyan oceanic subduction within central Qiangtang, northern Tibet. *Lithos*, *127*(1–2), 39–53. https://doi.org/10.1016/j.lithos.2011.07.023

Liu, Y., Tan, J., Wei, J. H., Zhao, S. Q., Liu, X. Y., Gan, J. R., & Wang, Z. H. (2019). Sources and petrogenesis of Late Triassic Zhiduo volcanics in the northeast Tibet: Implications for tectonic evolution of the western Jinsha Paleo-Tethys Ocean. *Lithos*, 169–182. https://doi.org/10.1016/j.lithos.2019.04.005

Liu, Y., Xiao, W., Windley, B. F., Schulmann, K., Li, R., Ji, W., et al. (2019). Late Silurian to Late Triassic seamount/oceanic plateau series accretion in Jinshajiang subduction mélange, Central Tibet, SW China. *Geological Journal*, *54*(2), 961–977. https://doi.org/10.1002/gj.3432

Ludwig, K. R. (2012). *User's Manual for isoplot 3.75: A Geochronological Toolkit for Microsoft Excel* (Vol. 5, pp. 1–75). Berkeley Geochronology Center Special Publication.

Ma, Y. M., Wang, Q., Wang, J., Yang, T. S., Tan, X. D., Dan, W., et al. (2019). Paleomagnetic constraints on the origin and drift history of the North Qiangtang terrane in the Late paleozoic. *Geophysical Research Letters*, *46*, 689–697. https://doi.org/10.1029/2018GL080964

McFadden, P. L. (1990). A new fold test for paleomagnetic studies. *Geophysical Journal International*, *103*, 163–169. https://doi.org/10.1111/j.1365-246x.1990.tb01761.x

McFadden, P. L., & Mcelhinny, M. W. (1990). Classification of the reversal test in paleomagnetism. *Geophysical Journal International*, *103*, 725–729. https://doi.org/10.1111/j.1365-246x.1990.tb05683.x

Meert, J. G., Pivarunas, A. F., Evans, D. A. D., Pisarevsky, S. A., Pesonen, L. J., Li, Z. X., et al. (2020). The magnificent seven: A proposal for modest revision of the quality index. *Tectonophysics*, *790*, 228549. https://doi.org/10.1016/j.tecto.2020.228549

Metcalfe, I. (1996). Gondwanaland dispersion, Asian accretion and evolution of eastern Tethys. *Australian Journal of Earth Sciences*, *43*(6), 605–623. https://doi.org/10.1080/08120099608728282

Metcalfe, I. (2013). Gondwana dispersion and Asian accretion: Tectonic and paleogeographic evolution of eastern Tethys. *Journal of Asian Earth Sciences*, *66*, 1–33. https://doi.org/10.1016/j.jseaes.2012.12.020

Pan, G. T., Zhu, D. C., Wang, L. Q., Liao, Z. L., Geng, Q. R., & Jiang, X. S. (2004). Bangong Lake-Nu River suture zone-the northern boundary of Gondwanaland: Evidence from geology and geophysics. *Earth Science Frontiers*, *11*, 371–382. (In Chinese).

Pullen, A., Kapp, P., Gehrels, G. E., Vervoort, J. D., & Ding, L. (2008). Triassic continental subduction in central Tibet and Mediterranean-style closure of the Paleo-Tethys Ocean. *Geology*, *36*, 351. https://doi.org/10.1130/g24435a.1



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Qinghai Geological Survey Institute (QGSI). (2005a). *Geological map (1:250000) of the People's Republic of China* (p. I46). Zhiduo County sheet.
Qinghai Geological Survey Institute (QGSI). (2005b). *Geological map (1:250000) of the People's Republic of China*. Zaduo County sheet.
Qinghai Geological Survey Institute (QGSI). (2014). *Geological Survey Report of the People's Republic of China-Zhiduo County sheet*. China University of Geosciences Press (ISBN: 9787562533849).
Reid, A. J., Wilson, C. J. L., & Liu, S. (2005). Structural evidence for the Permo-Triassic tectonic evolution of the Yidun Arc, eastern Tibetan Plateau. *Journal of Structural Geology*, *27*, 119–137. https://doi.org/10.1016/j.jsg.2004.06.011
Roger, F., Jolivet, M., & Malavieille, J. (2010). The tectonic evolution of the Songpan-Garzê (North Tibet) and adjacent areas from Proterozoic to Present: A synthesis. *Journal of Asian Earth Sciences*, *39*(4), 254–269. https://doi.org/10.1016/j.jseaes.2010.03.008
Sengör, A. M. C. (1979). Mid-Mesozoic closure of Permo-Triassic Tethys and its implications. *Nature*, *279*, 590–593.
Sha, J. G., & Fürsich, F. T. (1999). Paleotethys Ocean closed before capitanian times in Hohxil area (W China): New data on the temporal extend of the Paleotethys. *Neues Jahrbuch für Geologie und Paläontologie—Abhandlungen*, *7*, 440–448. https://doi.org/10.1127/njgpm/1999/1999/440
Song, C. Y., Wang, J., Fu, X. G., Feng, X. L., Chen, M., & He, L. (2012). Late Triassic paleomagnetic data from the Qiangtang terrane of Tibetan Plateau and their tectonic significances. *Journal of Jilin University (Earth Science Edition)*, *42*, 526–535. (In Chinese).
Song, P. P., Ding, L., Li, Z. Y., Lippert, P. C., Yang, T. S., Zhao, X. X., et al. (2015). Late Triassic paleolatitude of the Qiangtang block: Implications for the closure of the Paleo-Tethys Ocean. *Earth and Planetary Science Letters*, *424*, 69–83. https://doi.org/10.1016/j.epsl.2015.05.020
Song, P. P., Ding, L., Li, Z. Y., Lippert, P. C., & Yue, Y. H. (2017). An early bird from Gondwana: Paleomagnetism of Lower Permian lavas from northern Qiangtang (Tibet) and the geography of the Paleo-Tethys. *Earth and Planetary Science Letters*, *475*, 119–133. https://doi.org/10.1016/j.epsl.2017.07.023
Song, P. P., Ding, L., Lippert, P. C., Li, Z. Y., Zhang, L. Y., & Xie, J. (2020). Paleomagnetism of Middle Triassic lavas from northern Qiangtang (Tibet): Constraints on the closure of the Paleo-Tethys Ocean. *Journal of Geophysical Research: Solid Earth*, *125*, e2019JB017804. https://doi.org/10.1029/2019jb017804
Stampfli, G. M., von Raumer, J. F., & Borel, G. D. (2002). Paleozoic evolution of pre-Variscan terranes: From Gondwana to the Variscan collision. In J. R. Martínez Catalán, R. D. Hatcher, Jr., R. Arenas, & F. Díaz García (Eds.), *Variscan-Appalachian Dynamics: The Building of the Late Paleozoic Basement: Boulder. Colorado*. Geological Society of America Special Paper. (pp. 263–280). https://doi.org/10.1130/0-8137-2364-7.263
Tan, J., Wei, J. H., Zhao, S. Q., Li, Y. J., Liu, Y., Liu, X. Y., et al. (2020). Petrogenesis of Late Triassic high-Mg diorites and associated granitoids with implications for Paleo-Tethys evolution in the northeast Tibetan Plateau. *Bulletin*, *132*(5–6), 955–976. https://doi.org/10.1130/b35225.1
Tang, Y., Qin, Y. D., Gong, X. D., Duan, Y. Y., Chen, G., & Yao, H. Y. (2020). Discovery of eclogites in Jinsha River suture zone, Gonjo County, eastern Tibet and its restriction on Paleo-Tethyan evolution. *China Geology*, *3*, 83–103. https://doi.org/10.31035/cg2020003
Tibet Autonomous Geological Survey (TARGS) (2007). *Geological map (1:250000) of the People's Republic of China-Markam County Sheet*. China University of Geosciences Press.
Torsvik, T. H., Van der Voo, R., Preeden, U., Niocaill, C. M., Steinberger, B., & Doubrovine, P. V. (2012). Phanerozoic polar wander and paleogeography and dynamics. *Earth-Science Reviews*, *114*, 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007
Van der Voo, R. (1990). The reliability of paleomagnetic data. *Tectonophysics*, *184*, 1–9. https://doi.org/10.1016/0040-1951(90)90116-p
Wang, X., Metcalfe, I., Jian, P., He, L., & Wang, C. (2000). The Jinshajiang–Ailaoshan suture zone, China: Tectonostratigraphy, age and evolution. *Journal of Asian Earth Sciences*, *18*(6), 675–690. https://doi.org/10.1016/s1367-9120(00)00039-0
Watson, G. S., & Enkin, R. J. (1993). The fold test in paleomagnetism as a parameter estimation problem. *Geophysical Research Letters*, *20*(19), 2135–2137. https://doi.org/10.1029/93gl01901
Wu, L., Kravchinsky, V. A., & Potter, D. K. (2017). Apparent polar wander paths of the major Chinese blocks since the Late Paleozoic: Toward restoring the amalgamation history of east Eurasia. *Earth-Science Reviews*, *171*, 492–519. https://doi.org/10.1016/j.earscirev.2017.06.016
Yan, M. D., Zhang, D. W., Fang, X. M., Ren, H. D., Zhang, W. L., Zan, J. B., et al. (2016). Paleomagnetic data bearing on the Mesozoic deformation of the Qiangtang Block: Implications for the evolution of the Paleo- and Meso-Tethys. *Gondwana Research*, *39*, 292–316. https://doi.org/10.1016/j.gr.2016.01.012
Yan, Y. G., Huang, B. C., Zhao, J., Zhang, D. H., Liu, X. H., Charusiri, P., et al. (2017). Large southward motion and clockwise rotation of Indochina throughout the Mesozoic: Paleomagnetic and detrital zircon U–Pb geochronological constraints. *Earth and Planetary Science Letters*, *459*, 264–278. https://doi.org/10.1016/j.epsl.2016.11.035
Yan, Y. G., Zhao, Q., Zhang, D. H., Charusiri, P., Huang, B. C., & Zhang, P. Z. (2019). Palaeomagnetism of Late Triassic volcanic rocks from the western margin of Khorat Basin, Thailand and its implication for ambiguous inclination shallowing in Mesozoic sediments of Indochina. *Geophysical Journal International*, *219*, 897–910. https://doi.org/10.1093/gji/ggz331
Yan, Y. G., Zhao, Q., Zhang, Y. P., Huang, B. C., Zheng, W. J., & Zhang, P. Z. (2019). Direct paleomagnetic constraint on the closure of Paleo-Tethys and its implications for linking the Tibetan and Southeast Asian blocks. *Geophysical Research Letters*, *46*, 14368–14376. https://doi.org/10.1029/2019gl085473
Yang, T. N., Hou, Z. Q., Wang, Y., Zhang, H. R., & Wang, Z. L. (2012). Late Paleozoic to Early Mesozoic tectonic evolution of northeast Tibet: Evidence from the Triassic composite western Jinsha-Garzê-Litang suture. *Tectonics*, *31*(4). https://doi.org/10.1029/2011tc003044
Yang, X. F., Cheng, X., Zhou, Y. N., Ma, L., Zhang, X. D., Yan, Z. S., et al. (2017). Paleomagnetic results from Late Carboniferous to Early Permian rocks in the northern Qiangtang terrane, Tibet, China, and their tectonic implications. *Science China Earth Sciences*, *60*, 124–134. https://doi.org/10.1007/s11430-015-5462-7
Yuan, C., Zhou, M. F., Sun, M., Zhao, Y., Wilde, S., Long, X., & Yan, D. (2010). Triassic granitoids in the eastern Songpan Ganzi Fold Belt, SW China: Magmatic response to geodynamics of the deep lithosphere. *Earth and Planetary Science Letters*, *290*(3–4), 481–492. https://doi.org/10.1016/j.epsl.2010.01.005
Zhai, Q. G., Zhang, R. Y., Jahn, B. M., Li, C., Song, S. G., & Wang, J. (2011). Triassic eclogites from central Qiangtang, northern Tibet, China: Petrology, geochronology and metamorphic P-T path. *Lithos*, *125*, 173–189. https://doi.org/10.1016/j.lithos.2011.02.004
Zhang, H. F., Parrish, R., Zhang, L., Xu, W. C., Yuan, H. L., Gao, S., & Crowley, Q. G. (2007). A-type granite and adakitic magmatism association in Songpan–Garze fold belt, eastern Tibetan Plateau: Implication for lithospheric delamination. *Lithos*, *97*(3–4), 323–335. https://doi.org/10.1016/j.lithos.2007.01.002
Zhang, K. J., Cai, J. X., Zhang, Y. X., & Zhao, T. P. (2006). Eclogites from central Qiangtang, northern Tibet (China) and tectonic implications. *Earth and Planetary Science Letters*, *245*, 722–729. https://doi.org/10.1016/j.epsl.2006.02.025
Zhang, K. J., Zhang, Y. X., Li, B., Zhu, Y. T., & Wei, R. Z. (2006). The blueschist-bearing Qiangtang metamorphic belt (northern Tibet, China) as an in situ suture zone: Evidence from geochemical comparison with the jinsa suture. *Geology*, *34*, 493. https://doi.org/10.1130/G22404.1
Zhang, L. Y., Ding, L., Pullen, A., Xu, Q., Liu, D. L., Cai, F. L., et al. (2014). Age and geochemistry of western Hoh-Xil–Songpan-Ganzi granitoids, northern Tibet: Implications for the Mesozoic closure of the Paleo-Tethys Ocean. *Lithos*, *190–191*, 328–348. https://doi.org/10.1016/j.lithos.2013.12.019



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Zhang, N., Li, J. B., Yang, Y. S., & Na, F. C. (2012). Petrogeochemical characteristics and tectonic setting of the Wandaohu ophiolite mélange, Jinshajiang suture, Tibet. *Acta Petrologica Sinica*, *28*(4), 1291–1304. (In Chinese).

Zhang, Y. C., Shi, G. R., & Shen, S. Z. (2013). A review of Permian stratigraphy, palaeobiogeography and palaeogeography of the Qinghai–Tibet Plateau. *Gondwana Research*, *24*(1), 55–76. https://doi.org/10.1016/j.gr.2012.06.010

Zhao, G. C., Wang, Y. J., Huang, B. C., Dong, Y. P., Li, S. Z., Zhang, G. W., et al. (2018). Geological reconstructions of the East Asian blocks: From the breakup of Rodinia to the assembly of Pangea. *Earth-Science Reviews*, *186*, 262–286. https://doi.org/10.1016/j.earscirev.2018.10.003

Zhao, J., Huang, B. C., Yan, Y. G., & Zhang, D. H. (2015). Late Triassic paleomagnetic result from the Baoshan Terrane, West Yunnan of China: Implication for orientation of the East Paleotethys suture zone and timing of the Sibumasu-Indochina collision. *Journal of Asian Earth Sciences*, *111*, 350–364. https://doi.org/10.1016/j.jseaes.2015.06.033

Zhao, S. Q., Tan, J., Wei, J. H., Tian, N., Zhang, D. H., Liang, S. N., & Chen, J. J. (2014). Late Triassic Batang Group arc volcanic rocks in the northeastern margin of Qiangtang terrane, northern Tibet: Partial melting of juvenile crust and implications for Paleo-Tethys ocean subduction. *International Journal of Earth Sciences*, *104*(2), 369–387. https://doi.org/10.1007/s00531-014-1080-z

Zhao, X. M., Niu, Z. J., Yao, H. Z., Duan, Q. F., & Tang, C. Y. (2006). Division contrast and environment analysis of the Triassic Batang Group in Northwest Zhiduo area of Qinghai Province. *Acta Sedimentologica Sinica*, *24*, 511–520. (In Chinese).

Zhao, Z. Z., Li, Y. T., Ye, H. F., & Zhang, Y. E. (2001). *Stratigraphy in the Tibet plateau* (pp. 10–201). Science Press. (In Chinese).

Zhou, Y. N., Cheng, X., Wu, Y. Y., Kravchinsky, V., Shao, R. Q., Zhang, W. J., et al. (2019). The northern Qiangtang Block rapid drift during the Triassic Period: Paleomagnetic evidence. *Geoscience Frontiers*, *10*, 2313–2327. https://doi.org/10.1016/j.gsf.2019.05.003

Zhu, D. C., Zhao, Z. D., Niu, Y., Dilek, Y., Hou, Z. Q., & Mo, X. X. (2013). The origin and pre-Cenozoic evolution of the Tibetan Plateau. *Gondwana Research*, *23*(4), 1429–1454. https://doi.org/10.1016/j.gr.2012.02.002

Zi, J. W., Cawood, P. A., Fan, W. M., Tohver, E., Wang, Y. J., & McCuaig, T. C. (2012). Generation of Early Indosinian enriched mantle-derived granitoid pluton in the Sanjiang Orogen (SW China) in response to closure of the Paleo-Tethys. *Lithos*, *140–141*, 166–182. https://doi.org/10.1016/j.lithos.2012.02.006

## References From the Supporting Information

Deenen, M. H. L., Langereis, C. G., van Hinsbergen, D. J. J., & Biggin, A. J. (2014). Erratum: Geomagnetic secular variation and the statistics of palaeomagnetic directions. *Geophysical Journal International*, *197*, 643.

Dunlop, D. J., & Özdemir, Ö. (1997). *Rock magnetism: Fundamentals and frontiers*. Cambridge University Press.

Harrison, R. J., & Feinberg, J. M. (2008). FORCinel: An improved algorithm for calculating first-order reversal curve distributions using locally weighted regression smoothing. *Geochemistry, Geophysics, Geosystems*, *9*, Q05016.

Johnson, C. L., Constable, C. G., Tauxe, L., Barendregt, R., Brown, L. L., Coe, R. S., et al. (2008). Recent investigations of the 0-5 Ma geomagnetic field recorded by lava flows. *Geochemistry, Geophysics, Geosystems*, *9*, Q04032.

Jones, C. H. (2002). User-driven integrated software lives: "PaleoMag" paleomagnetics analysis on the Macintosh. *Computers & Geosciences*, *28*(10), 1145–1151.

Kodama, K. P. (2012). *Paleomagnetism of Sedimentary Rocks: Process and Interpretation*. John Wiley & Sons.



19448007, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021GL096902 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License